Does prediction during disfluency boost memory for predicted meanings?

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Disfluency is a common[1] feature of speech that listeners use as a meaningful cue[2] to predict upcoming meanings. When listening to a disfluent repair (e.g., "*salt… uh I mean*…"), listeners generate predictions during the reparandum (*salt*) and editing period (*uh, I mean*), considering plausible alternatives (*pepper*)[3]. The reparandum itself lingers in memory[4] and influences later processing[5]. Studies of temporary ambiguity in language processing likewise reveal predicted—but unrealized—meanings are retained in memory[5,6], sometimes resulting in false recognition for predicted words[7], particularly when the period of ambiguity is prolonged[8]. Unknown, however, is whether this disfluency-driven prediction process boosts memory for predicted items above and beyond the memory boost that would be expected simply due to strong semantic association[9] creating an illusion of memory[10] for predicted items (salt -> pepper).

A recent study[4] demonstrated that items predicted during the reparandum are boosted in memory relative to unrelated items (e.g., milk). However, this design cannot distinguish a memory boost due to semantic association[10], from a memory boost due to prediction of the plausible alternative (e.g., *salt, uh….* → pepper). We test the hypothesis that prediction during the editing period of a disfluent repair (*uh, I mean*) boosts memory for predicted items, above and beyond any memory boost due to semantic association by comparing memory for predicted items to distractors semantically related to the mentioned repair (e.g., mustard) as in [1]:

[1] *… the chef reached for some salt uh I mean some ketchup…*

If disfluency-driven prediction boosts memory for predicted items, we should observe a memory boost for predicted alternatives relative to the distractor, particularly when the editing period is long. Prediction is weaker for coordination constructions[3] (*salt and…*) which serve as control.

Method. Participants (*Ps*) viewed scenes with four pictures (**Fig1**) and heard a related sentence in English which mentioned two items, each of which appeared with a semantically related item on screen (e.g., salt, pepper, ketchup, mustard). Sentences contained a repair or a coordination phrase, and contained short or long periods during which participants could make a prediction about what would be named. In total, *Ps* (N = 64, pre-registered) listened to 40 critical sentences (with 16 interspersed fillers), including repairs (short: "uh"; long: "uh, I mean") or coordination (short: "and"; long: "as well as"). A surprise 2AFC memory test probed memory for all 160 images; *Ps* saw two images (one old, one new) and were asked to click the old one. **Results**. A mixed-effects logistic regression analysis was fit in glmer with random effects determined by buildmer; fixed effects were contrast coded including Helmert contrasts for item type (repair, reparandum, repair-related, reparandum-related) with binary memory accuracy as dependent (**Fig2**). As predicted, the repair (e.g., ketchup) was remembered better than the reparandum and distractors ($\beta = -0.13054$, $p = .0145$), and the reparandum (e.g., salt) was remembered better than the semantically related items ($\beta = -0.18253$, $p = .0012$). Critically, the unmentioned item that could have been predicted (e.g., *"…salt uh I mean…"* → *pepper*) was not better remembered than the unmentioned item that was unlikely to be predicted (i.e., memory for reparandum-related and repair-related items was similar, $\beta = 0.033$, $p = .6052$). Effects of sentence form (coordination vs. disfluency) and editing period length were not significant.

Conclusion. Listeners make predictions about the speaker's intended message during disfluent repairs[3], yet previously unknown was whether prediction causes the predicted meanings to linger in memory above and beyond any memory boost due to the semantic association between the reparandum and the (incorrectly) predicted repair. We observed a clear boost to memory for mentioned (reparandum, repair) over unmentioned semantically related items. However, there was no evidence that predictive processes boosted the representation of the predicted items in memory. In sum, while it is clear that language-driven activation of referential candidates boosts memory for them[7-8,11], these findings provide no evidence that *prediction* per se amplifies this process. These findings are consistent with models which posit that the discourse history is a context-marked probabilistic memory representation including all candidate meanings, regardless of whether they were realized in speech, or not[8].

Figure1: Example study trial. Participants saw 4 images: the reparandum (e.g., *salt*), reparandum-related (e.g., *pepper*), the repair (e.g., *ketchup*), and a repair-related (e.g., *mustard*). We counterbalanced across lists which version of the sentence participants heard:

*The meat was pretty bland, so the chef reached for some salt **uh I mean** some ketchup/salt **uh** some ketchup/salt **as well as** some ketchup/salt **and** some ketchup, which made it much more flavorful.*

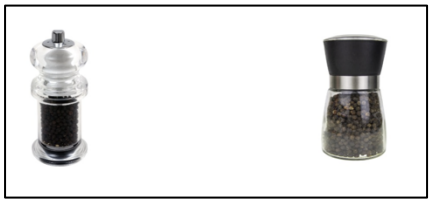

Figure 2: Example 2AFC memory trial. Participants were shown an old image (left, presented during the reference task) alongside a similar, new image (right) and were asked to click on the image they had seen in the language task (which items were old/ new was counterbalanced across lists).

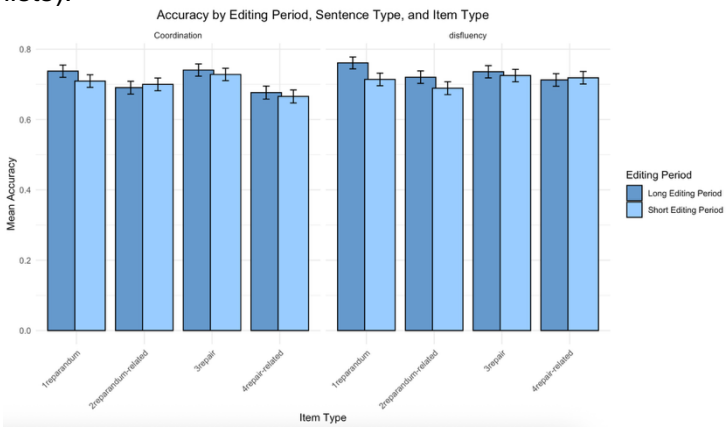

Figure 3: Accuracy on the 2AFC memory test by item type and condition.

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